

Database Schema

Overview

Attribute	Value
Database	PostgreSQL
ORM	Prisma
Schema source	backend/prisma/schema.prisma
Documentation link	https://docs.google.com/document/d/1F...

Entity Relationship Diagram

[Diagram](#) - Here you can find the diagram

Enums

These enums are defined in the database (PostgreSQL enum types) and used by multiple tables.

Enum	Values	Used in
Locale	en, ru	disease_translation, symptom_translat...
SymptomCategory	sign, symptom	Symptom.category
SymptomPriority	primary, secondary	DiseaseSymptom.priority
Typicality	typical, possible, rare	DiseaseSymptom.typicality
RiskDirection	risk, protective	DiseaseRiskFactor.direction

Tables

Disease

Stores the primary disease entities.

Column	Type	Constraints	Description
disease_id	SERIAL	PK	Internal identifier
code	VARCHAR(10)	UNIQUE, NOT NULL	Disease code (documented as ICD-10 in...

name	TEXT	NOT NULL	Disease name (default locale)
description	TEXT	NULLABLE	Disease description
prevention	TEXT	NULLABLE	Prevention recommendations

Relations: - 1:N to DiseaseSymptom - 1:N to DiseaseRiskFactor - 1:N to DiseaseTranslation

Symptom

Stores standardized symptoms/signs.

Column	Type	Constraints	Description
symptom_id	SERIAL	PK	Internal identifier
code	VARCHAR(10)	UNIQUE, NULLABLE	Optional code (used in API docs as ?S...
term	TEXT	UNIQUE, NOT NULL	Canonical symptom term (default locale)
category	SymptomCategory	NULLABLE	sign or symptom

Relations: - 1:N to DiseaseSymptom - 1:N to SymptomTranslation

RiskFactor

Stores standardized risk factors and protective factors.

Column	Type	Constraints	Description
risk_factor_id	SERIAL	PK	Internal identifier
code	VARCHAR(10)	UNIQUE, NULLABLE	Optional code
name	TEXT	NOT NULL	Risk factor name (default locale)
definition	TEXT	NULLABLE	Definition/description

Relations: - 1:N to DiseaseRiskFactor - 1:N to RiskFactorTranslation

DiseaseSymptom

Join table connecting diseases and symptoms, with extra attributes for clinical interpretation.

Column	Type	Constraints	Description
disease_symptom_id	SERIAL	PK	Internal identifier
disease_id	INTEGER	FK ? Disease(disease_id), NOT NULL	Disease reference

symptom_id	INTEGER	FK ? Symptom(symptom_id), NOT NULL	Symptom reference
priority	SymptomPriority	NOT NULL	primary or secondary
typicality	Typicality	NOT NULL, DEFAULT typical	typical, possible, or rare

Constraints / Indexes: - UNIQUE(disease_id, symptom_id) - INDEX(disease_id) name: idx_disease_symptoms_d - INDEX(symptom_id) name: idx_disease_symptoms_s

DiseaseRiskFactor

Join table connecting diseases and risk factors, with direction (risk vs protective).

Column	Type	Constraints	Description
disease_risk_factor_id	SERIAL	PK	Internal identifier
disease_id	INTEGER	FK ? Disease(disease_id), NOT NULL	Disease reference
risk_factor_id	INTEGER	FK ? RiskFactor(risk_factor_id), NOT NULL	RiskFactor reference
direction	RiskDirection	NOT NULL, DEFAULT risk	risk or protective

Constraints / Indexes: - UNIQUE(disease_id, risk_factor_id) - INDEX(disease_id) name: idx_disease_risk_d - INDEX(risk_factor_id) name: idx_disease_risk_rf

Source

Stores dataset / organization sources.

Column	Type	Constraints	Description
source_id	SERIAL	PK	Internal identifier
name	TEXT	UNIQUE, NOT NULL	Organization/source name
link	TEXT	UNIQUE, NOT NULL	Source URL

disease_translation

Localized fields for Disease per locale.

Column	Type	Constraints	Description
id	SERIAL	PK	Internal identifier
disease_id	INTEGER	FK ? Disease(disease_id), NOT NULL	Disease reference
locale	Locale	NOT NULL	en or ru

name	TEXT	NULLABLE	Localized disease name
description	TEXT	NULLABLE	Localized description
prevention	TEXT	NULLABLE	Localized prevention

Constraints: - UNIQUE(disease_id, locale)

symptom_translation

Localized fields for Symptom per locale.

Column	Type	Constraints	Description
id	SERIAL	PK	Internal identifier
symptom_id	INTEGER	FK ? Symptom(symptom_id), NOT NULL	Symptom reference
locale	Locale	NOT NULL	en or ru
term	TEXT	NOT NULL	Localized term

Constraints: - UNIQUE(symptom_id, locale)

risk_factor_translation

Localized fields for RiskFactor per locale.

Column	Type	Constraints	Description
id	SERIAL	PK	Internal identifier
risk_factor_id	INTEGER	FK ? RiskFactor(risk_factor_id), NOT NULL	RiskFactor reference
locale	Locale	NOT NULL	en or ru
name	TEXT	NULLABLE	Localized name
definition	TEXT	NULLABLE	Localized definition

Constraints: - UNIQUE(risk_factor_id, locale)

Relationships

Relationship	Type	Description
Disease ? DiseaseSymptom	One-to-Many	A disease has many symptom links (wit...
Symptom ? DiseaseSymptom	One-to-Many	A symptom can be linked to many diseases

Disease ? DiseaseRiskFactor	One-to-Many	A disease has many risk-factor links ...
RiskFactor ? DiseaseRiskFactor	One-to-Many	A risk factor can be linked to many d...
Disease ? disease_translation	One-to-Many	Localized disease fields per locale
Symptom ? symptom_translation	One-to-Many	Localized symptom term per locale
RiskFactor ? risk_factor_translation	One-to-Many	Localized risk factor fields per locale

Migrations

Migrations are stored in `backend/prisma/migrations/`.

Version	Folder	Description
001	20251125210618_diseasesdb	Initial schema: diseases, symptoms, r...

Seeding

Run migrations and seed data from the backend folder:

```
# from backend/
npm run migrate
npm run generate
npm run seed
```

Notes: - `npm run seed` runs `ts-node prisma/seed.ts`. - The seed script populates diseases, symptoms, risk factors, and sources (see `backend/prisma/seeder/`).